

Amal Chakhar, PhD

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Profile

Earth-observation scientist (PhD) with 10 peer-reviewed papers, applying machine learning to Sentinel-1/-2 and Landsat imagery for agriculture. Builds end-to-end pipelines — from ingestion and feature engineering to classification, validation and policy-ready maps — to answer real questions about crops, water and drought.

Skills

Programming: Python (pandas, NumPy, scikit-learn), JavaScript (Google Earth Engine), R, Git.
Earth Observation: Sentinel-1 (SAR), Sentinel-2, Landsat-8, UAV photogrammetry, time-series analysis, datacubes.
Machine Learning: Supervised classification (Random Forest, SVM, kNN), accuracy assessment, feature engineering on multi-source EO data.
Geospatial: Google Earth Engine, QGIS, ArcGIS, GDAL.
Domain: Crop classification, soil moisture retrieval, irrigation detection, evapotranspiration, drought monitoring.
Languages: Spanish, English, French (professional working proficiency); Arabic (native).

Experience

- Research Fellow, Remote Sensing & Crop Modelling**, University of Leeds – Leeds, UK Jul 2024 – Present
- Build multi-sensor Earth observation pipelines (Sentinel-1/-2 + machine learning) for the iSPARK programme — crop classification and drought monitoring across smallholder farming systems in western Kenya to support food-security policy.
 - Mapping drought stress in maize monocrop vs. intercrop systems; presented at the International Crop Modelling Symposium (iCROP2026), Florence, 2026.
- Postdoctoral Researcher**, University of Castilla-La Mancha – Albacete, Spain Mar 2023 – Jun 2024
- Applied UAVs and optical/radar satellite fusion to vine disease detection and irrigation monitoring for precision-agriculture applications.
- Doctoral Researcher**, University of Castilla-La Mancha – Albacete, Spain 2019 – 2023
- Benchmarked 22 non-parametric classifiers across 12 crop types and 2,000+ field plots in semi-arid Spain, fusing Landsat-8 and Sentinel-2 to reach **~90% F1** (*Remote Sensing*, 2020).
 - Boosted crop-classification accuracy by integrating Sentinel-1 SAR (VV/VH) with Sentinel-2 NDVI across 9 feature scenarios, identifying the optimal feature–classifier pairing (*Remote Sensing*, 2021).
 - Detected irrigated vs. rainfed orchards at **88.4% accuracy** from Sentinel-1/-2 time series, and built an automated Sentinel Hub download-and-tiling pipeline (*Remote Sensing*, 2024).
 - Retrieved surface soil moisture (**R² up to 0.78**) by inverting IEM+WCM scattering models with neural networks, using UAV photogrammetry to measure surface roughness (*Remote Sensing*, 2021).

Selected Publications

[1] **Amal Chakhar** et al. “Irrigation Detection Using Sentinel-1 and Sentinel-2 Time Series on Fruit Tree Orchards”. In: *Remote Sensing* 16.3 (2024), p. 458. DOI: 10.3390/rs16030458.

[2] **Amal Chakhar** et al. “Improving the Accuracy of Multiple Algorithms for Crop Classification by Integrating Sentinel-1 Observations with Sentinel-2 Data”. In: *Remote Sensing* 13.2 (2021), p. 243. DOI: 10.3390/rs13020243.

[3] **Amal Chakhar** et al. “Assessing the Accuracy of Multiple Classification Algorithms for Crop Classification Using Landsat-8 and Sentinel-2 Data”. In: *Remote Sensing* 12.11 (2020), p. 1735. DOI: 10.3390/rs12111735.

Education

PhD, Agricultural and Environmental Sciences, University of Castilla-La Mancha, Spain 2023
MSc, Integrated Planning for Rural Development and Environmental Management, CIHEAM Zaragoza / Univ. Lleida, Spain 2018
Engineering Degree, Horticultural Systems, University of Sousse, Tunisia 2015